

CUSTOMER SHOPPING BEHAVIOR ANALYSIS

Uncovering insights from **3,900 transactions** across product categories — analyzing spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.



Dataset at a Glance

3,900

Transactions

18

Features

37

Missing Ratings

Key Feature Groups:

- **Demographics:** Age, Gender, Location, Subscription Status
- **Purchase Details:** Item, Category, Amount, Season, Size, Color
- **Behavior:** Discounts, Promo Codes, Frequency, Review Rating, Shipping

Data Preparation in Python

01

Load & Explore

Imported dataset with `pandas`; used `df.info()` and `.describe()` for structure and summary stats.

02

Clean & Standardize

Imputed missing Review Ratings with category median; renamed columns to `snake_case`; dropped redundant `promo_code_used`.

03

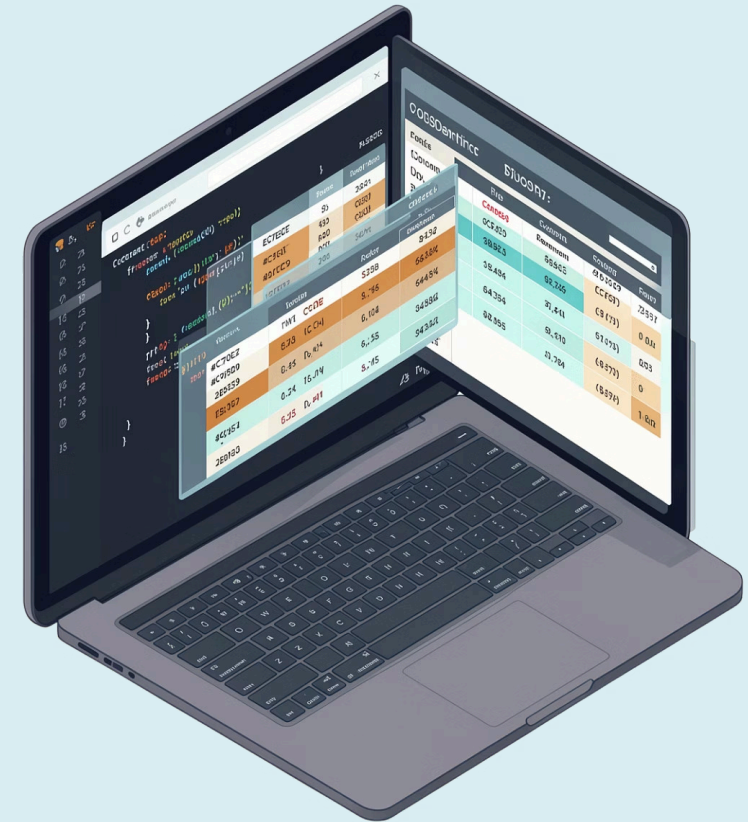
Feature Engineering

Created `age_group` bins and `purchase_frequency_days` from purchase data.

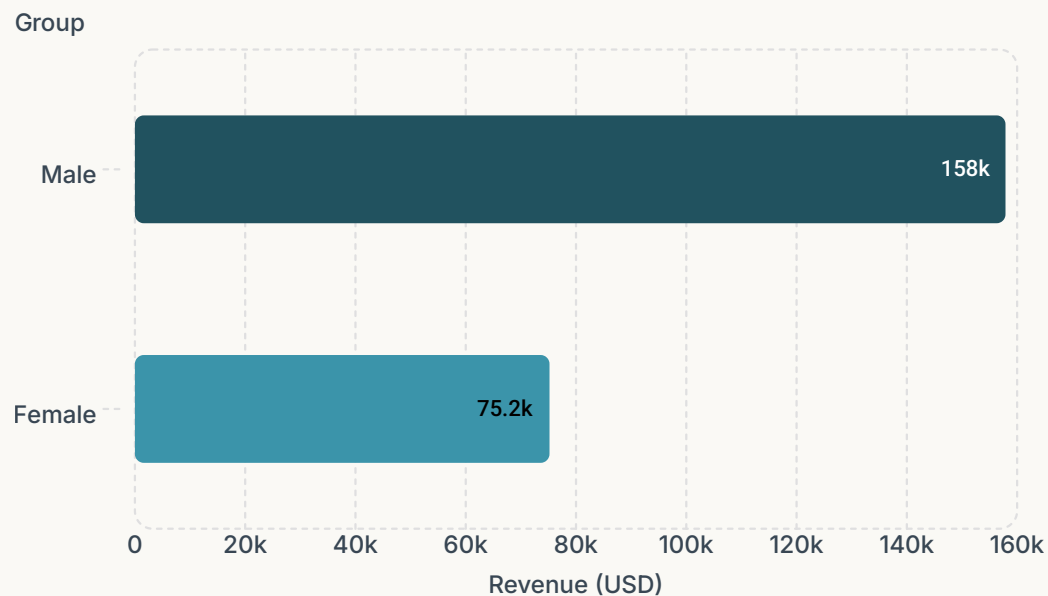
04

Database Integration

Loaded cleaned DataFrame into **PostgreSQL** for structured SQL analysis.



Revenue & Spending Insights



Key Findings

- Male customers generated **\$157,890** vs. female customers' **\$75,191** in total revenue
- **839 customers** used discounts yet still spent above the average purchase amount
- Express shipping users averaged **\$60.48** vs. Standard at **\$58.46**

Top-Rated Products

Top 5 by Average Rating

Product	Avg Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Most Discount-Dependent Products

Product	Discount Rate
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

Top Products by Category

Accessories

- Jewelry — 171
- Sunglasses — 161
- Belt — 161

Clothing

- Blouse — 171
- Pants — 171
- Shirt — 169

Footwear

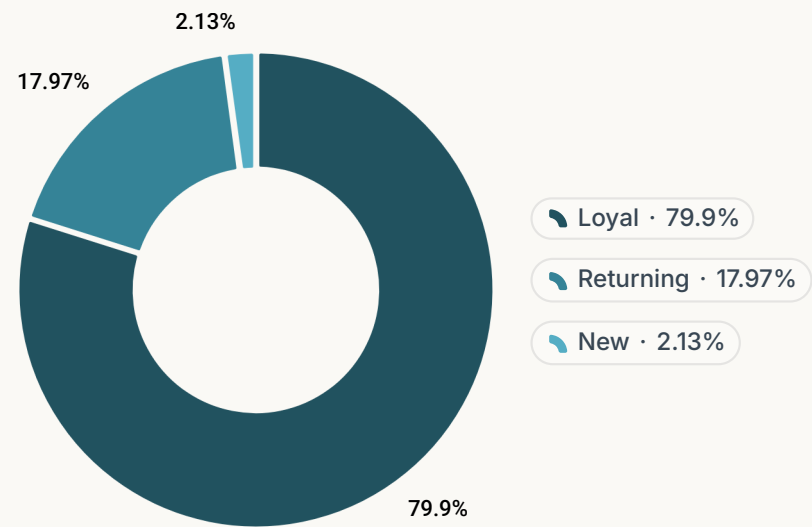
- Sandals — 160
- Shoes — 150
- Sneakers — 145

Outerwear

- Jacket — 163
- Coat — 161



Customer Segmentation

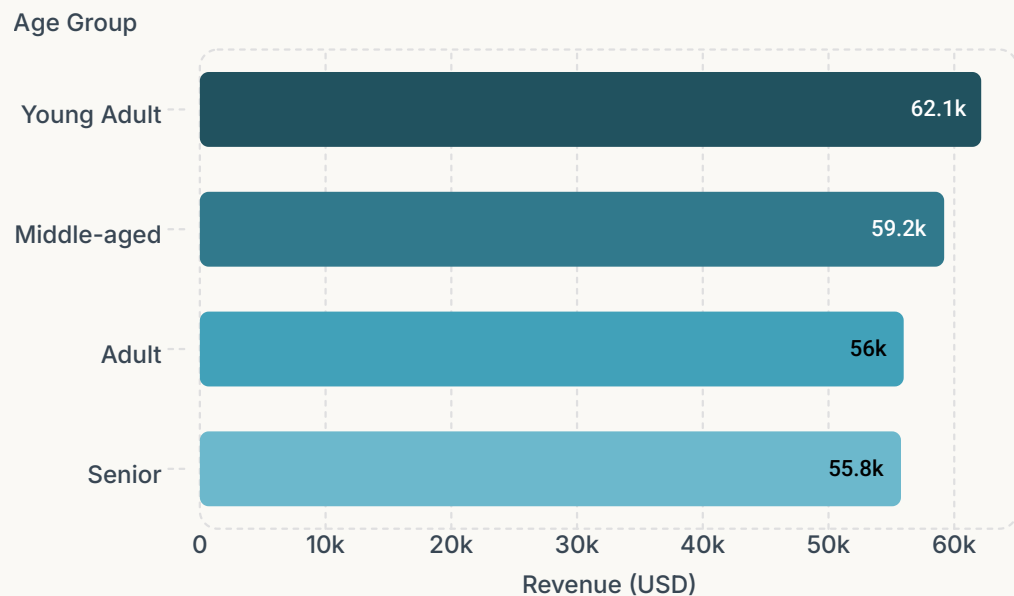


Subscription vs. Non-Subscribers

Status	Customers	Avg Spend	Total Revenue
Subscriber s	1,053	\$59.49	\$62,645
Non-Subscriber s	2,847	\$59.87	\$170,436

Customers with **>5 purchases**: 2,518 non-subscribers vs. 958 subscribers — repeat buyers skew non-subscribed.

Revenue by Age Group



Takeaway

Young Adults lead revenue at **\$62,143**, followed closely by Middle-aged customers. Revenue is relatively balanced across all age groups, suggesting broad market appeal with a slight skew toward younger shoppers.



Power BI Dashboard

3.9K

Customers

\$59.76

Avg Purchase

3.75

Avg Rating

Interactive filters by Subscription Status, Gender, Category, and Shipping Type. Visualizations include revenue & sales by category, age group breakdowns, and subscription distribution (27% Yes / 73% No).

Business Recommendations



Boost Subscriptions

Promote exclusive benefits to convert non-subscribers.



Loyalty Programs

Reward repeat buyers to accelerate movement into the Loyal segment.



Review Discount Policy

Balance sales boosts with margin control — especially for Hat, Sneakers, and Coat.



Targeted Marketing

Focus on high-revenue age groups and top-rated products in campaigns.